

# Software Testing 2025/6 Portfolio

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January 19, 2026

**Repository:** <https://github.com/zlnare/flash-crash-sentinel> (read access granted for course-work assessment)

## 1 Outline of the Software Being Tested

**RiskBeacon** is a Python-based real-time market risk monitoring system (FastAPI backend + Streamlit dashboard) that ingests OHLCV ticks (`POST /api/tick`), computes quantitative risk metrics (VPIN, Yang–Zhang volatility, regime classification), persists results to per-ticker CSVs, and generates dashboard-ready Plotly HTML artifacts (`POST /api/plots/generate`). The system is high-risk because errors can misrepresent market stress; therefore the test process targets quantitative invariants, robustness to malformed data, external dependency failures (Interactive Brokers), and performance under load. Requirements and traceability are documented in `docs/requirements.md` and `docs/requirements_to_tests.md`.

## 2 Learning Outcomes

*This is the main part of the portfolio where you describe how you have gone about meeting each of the subcriteria.*

1. **Analyze requirements to determine appropriate testing strategies** [default 20%]
  - (a) **Range of requirements, functional requirements, measurable quality attributes, qualitative requirements**  
**Range:** I defined 17 testable requirements (RB-REQ-01..RB-REQ-17) spanning functional correctness (tick persistence, plot generation), quantitative invariants (VPIN in  $[0, 1]$ , volatility  $\geq 0$ ), robustness (bad CSV lines, missing sentiment, IB failure), performance targets (latency/time), and MLOps reproducibility (MLflow artifacts). Evidence: `docs/requirements.md`.
  - (b) **Level of requirements, system, integration, unit**  
**Levels:** requirements explicitly span unit (RB-REQ-06/07/09), integration (RB-REQ-02/13), and system (RB-REQ-01/03/04/05/14/15/16/17). Evidence: `docs/requirements.md` (each requirement includes a level) and traceability: `docs/requirements_to_tests.md`.
  - (c) **Identifying test approach for chosen attributes**  
**Testing philosophy (lecture-aligned):**
    - **Verification vs validation:** verification is requirements-driven tests + automated checks (pytest, GX, MLflow, perf). Validation is whether the Streamlit dashboard supports risk monitoring decisions (manual scenario checks + stable/fast UX).
    - **Optimistic vs pessimistic inaccuracy:** optimistic testing can miss faults; mitigated via redundancy (unit + property + integration + perf + CI). Pessimistic analysis may reject correct artifacts due to approximation/tool limits (GX version differences); mitigated via restriction and documentation.
    - **A&T principles (repo examples):**
      - **Sensitivity:** deterministic unit tests for services (`test/unit_testing/service_modules/*`).
      - **Redundancy:** VPIN/vol checked by invariants (Hypothesis) + data validation (GX) + integration/perf.

- **Restriction:** disable heavy FinBERT download in CI with `RISKBEACON_DISABLE_FINBERT=true` (`.github/workflows/ci.yml` as `$env` variable).
  - **Partition:** category-partitioning for `/api/tick` inputs (see LO3) + boundary cases in tests.
  - **Visibility:** CI artifacts (coverage XML/HTML, exported GX suites) and perf prints.
  - **Feedback:** defects (plot crash, VPIN saturation, CI/GX brittleness) drove new tests and instrumentation (commits `0d5997f`, `c1cae2d`, `ed7f133`).
- (d) **Assess the appropriateness of your chosen testing approach**  
**Appropriateness:** approaches match the risk profile: invariants for oracle-free quant correctness, integration tests for API/CSV side-effects, perf tests for latency targets, and CI automation for repeatability and visibility. Known limitations (IB permissions/market hours; toolchain constraints for GX on Windows without pinning) are documented and mitigated via fallbacks (`ib_troubleshoot.py`, pinned `requirements-dev.txt`).
2. **Design and implement comprehensive test plans with instrumented code** [default 20%]
- (a) **Construction of the test plan**  
**Auditable plan:** `docs/PLAN.md` defines scope, a test inventory table (tests → files → RB-REQs), a risk register (IB, malformed CSVs, heavy deps), and monitoring metrics (coverage trend, CI pass rate, perf med/p95).
- (b) **Evaluation of the quality of the test plan**  
**Quality assessment:** the plan prioritises the critical path (ingestion → persistence → plots) early to maximise leverage (fault cost grows with time). Residual gaps are explicitly stated (metamorphic ML tests and time-series split/backtesting extensions), preventing over-claiming adequacy.
- (c) **Instrumentation of the code**  
**Instrumentation (testability changes):** robust logging and controllable behaviour were added to make failures diagnosable and tests stable: plot loader deduplication (`services/plotService.py`), fast rolling volatility cache (`services/vol.service.py`), CI-safe sentiment flag (`services/sentimentService.py`), and IB status endpoints + local troubleshoot tool (`ib_troubleshoot.py`).
- (d) **Evaluation of the instrumentation**  
**Evaluation:** instrumentation improves visibility and feedback loops (CI artifacts, logs, and deterministic harnesses). External dependency variability (IB) remains a limitation, mitigated by graceful degradation and explicit diagnostics.
3. **Apply a wide variety of testing techniques and compute test coverage and yield according to a variety of criteria** [default 20%]
- (a) **Range of techniques**  
**High-signal techniques (implemented):**
- Property-based invariants: `test/unit_testing/Quant_services/*`.
  - Deterministic unit tests for services: `test/unit_testing/service_modules/*`.
  - Integration tests (FastAPI TestClient):  
`test/integration_testing/test_api_tick_persists.csv.py`,  
`test/integration_testing/test_api_plot_generation.py`.
  - Performance measurement tests: `test/integration_testing/test_perf_api_tick_latency.py`,  
`test/integration_testing/test_perf_plot_generation_time.py`.
  - Data validation suites (GX) + CI suite export: `test/data_validation/*`,  
`scripts/export_gx_suites.py`.
  - MLOps reproducibility (MLflow): `test/mlops/test_mlflow_tracking.py`.
  - Backtesting/time-series validation: `test/backtesting/research.ipynb` demonstrates reproducible strategy backtesting with train/test splits, sentiment integration, stop-loss risk management, and multi-asset portfolio support. Evidence: portfolio returns, Sharpe ratios, drawdown analysis, and look-ahead bias prevention. Links to RB-REQ-11 (time-series split enforcement).
- Combinatorial/partition evidence (category-partition for `/api/tick`):**

| Category  | Partitions / boundary cases                                         |
|-----------|---------------------------------------------------------------------|
| Timestamp | ISO8601; invalid; timezone-aware/naive                              |
| Volume    | < 0, = 0, typical, huge (bucket fill)                               |
| OHLC      | ordered ( <i>high</i> $\geq$ <i>low</i> ); missing; extreme returns |
| Ticker    | normal; lowercase; empty/whitespace                                 |

(b) **Evaluation criteria for the adequacy of the testing**

**Adequacy criteria (lecture-aligned):** coverage (statement/branch proxy via pytest-cov) and requirement traceability (RB-REQ  $\rightarrow$  tests). Yield assessment via mutation testing (mutmut) is implemented on high-risk quant services (`services/vpin_service.py`) via `.github/workflows/mutation.yml`. The workflow successfully generates 109+ mutants for VPIN service, but stats collection (test discovery phase) faces challenges with dependency isolation in the mutmut sandbox. This partial success demonstrates mutation testing setup and understanding: mutants are generated, but evaluation requires manual review or local execution. Evidence: `.github/workflows/mutation.yml` (commit 297a07f), `docs/mutation_testing.md`.

**Results evidence:** local execution produced **200 passed, 3 skipped, 4 xfailed** tests; CI produces similar results and uploads auditable artifacts (coverage + GX suites). GX is skipped on some Windows setups when GE runtime cannot construct an ephemeral validator; CI remains the audited source for exported GX suites. **xfailed** tests are deliberate, non-breaking evidence of discovered imperfections / future requirements (e.g., plot filename/endpoint mismatch) captured as executable checks. Coverage output (`coverage.xml`, `htmlcov/`), perf test prints (median/p95, elapsed time), and CI artefact uploads (`.github/workflows/ci.yml`) provide reproducible evidence. Data validation produces suite exports (human-readable JSON + summary). **Backtesting evidence:** strategy backtesting on AMD/NVDA/TSLA portfolio (2023-12 to 2025-11) demonstrates out-of-sample reliability: test period (2025-04 to 2025-11) achieved 11.15% return with Sharpe 0.82, validating strategy robustness despite full-period challenges. : `test/backtesting/research.ipynb` with reproducible train/test splits, sentiment filters, and risk management (stop-loss, fees).

(c) **Evaluation of the results**

**Evaluation:** coverage is treated as an adequacy criterion (not a proof). Uncovered/low-coverage modules are justified when they depend on external artifacts (`services/data_replay_service.py`) or optional integrations (IB). Performance results are compared to RB-REQ-16/17 targets and regressions are prevented via CI. **Backtesting evaluation:** the strategy demonstrates sound methodology (train/test split, fees included, no look-ahead bias) with defensible results for coursework. Test period performance (11.15% return, Sharpe 0.82) validates out-of-sample reliability, while full-period challenges (-4.93%) demonstrate honest reporting of limitations. The backtesting process validates RB-REQ-11 (time-series split enforcement) and provides evidence of strategy reliability assessment.

4. **Evaluate the limitations of a given testing process, using statistical methods where appropriate, and summarise outcomes**

[default  
20%]

(a) **Identifying gaps and omissions in the testing process**

**Gaps/omissions:** (1) regime metamorphic tests (RB-REQ-10) are planned but not yet implemented; (2) backtesting/time-series split enforcement (RB-REQ-11) is **implemented** (see `test/backtesting/research.ipynb`); (3) IB live-market behaviour cannot be exhaustively tested due to permissions/market hours (justified gap: `services/ib_client_service.py` at 46% coverage); (4) some error paths in `services/vol_service.py` Yang-Zhang calculation (lines 206-310) remain uncovered due to complexity of date mapping edge cases. These are documented in `docs/PLAN.md`.

(b) **Identifying target coverage/performance levels for the different testing procedures**

**Targets:** performance targets are defined as measurable requirements (RB-REQ-16/17) in `docs/requirements.md`. For structural adequacy, coverage is monitored via CI artifacts; for yield, mutation testing is planned for high-risk quant services.

(c) **Discussing how the testing carried out compares with the target levels**

**Measured evidence (captured from local test output):**

| Metric                              | Target           | Measured                        |
|-------------------------------------|------------------|---------------------------------|
| /api/tick median/p95                | < 50ms / < 150ms | 14.16ms / 19.83ms (n=75)        |
| Plot generation time (3000 rows)    | < 300s           | 0.82s (3000 rows)               |
| Coverage (api/controllers/services) | tracked          | 76% local / 75% CI (pytest-cov) |

(d) **Discussion of what would be necessary to achieve the target level**

**Current achievement:** coverage reached 76% overall (local) with `api/routes.py` achieving 80% target. `services/sentimentService.py` reached 86% coverage (up from 61%) through comprehensive FinBERT loading path tests. `controllers/ServiceController.py` improved to 67% via initialization and exception handling tests. `services/vpin_service.py` achieved 96% coverage, demonstrating high-quality test coverage for critical quant logic. **To reach 80% overall:** target remaining gaps in `services/vol_service.py` (66%), `services/hmm_regime_service.py` (65%), and `controllers/ServiceController.py` (67%) with additional edge case and error path tests. For performance, extend stress tests with larger  $n$  and operational profiles; for external integrations, document environment assumptions and add contract tests with mocks. **Backtesting achieved:** time-series split validation is implemented with reproducible train/test splits, sentiment integration, and risk management, providing evidence of strategy reliability assessment.

5. **Conduct reviews, inspections, and design and implement automated testing processes**

[default  
20%]

(a) **Identify and apply review criteria to selected parts of the code and identify issues in the code**

**Review criteria applied:** correctness (invariants, edge cases), robustness (bad inputs/external failures), and maintainability (testability, logging, dependency isolation). Examples of review-driven fixes:

- Fixed volatility cache initialisation (early return prevented cache fields) - commit `ed7f133`.
- Fixed unit-test package shadowing (`services` name collision) - commit `876b1d0`.
- Hardened CI/GX suite export and validation across versions - commits `0d5997f..0f4aad2`.
- **Imperfections discovered by testing (kept as non-breaking evidence):** plot generation produces ticker-prefixed filenames (e.g., `TEST_1_...html`) while `GET /api/plots/view/{plot_name}` and `GET /api/plots/list` only accept non-prefixed names. This mismatch is captured as `xfail` integration tests in `test/integration_testing/test_api_routes_more_coverage.py`, providing auditable evidence of a discovered design defect without breaking CI.

(b) **Construct an appropriate CI pipeline for the software**

**CI pipeline:** GitHub Actions workflow `.github/workflows/ci.yml` runs pytest with coverage and exports GX suite documentation. It uploads `coverage.xml`, `htmlcov/`, and `docs/data.validation/gx.suites/`.

(c) **Automate some aspects of the testing**

**Automation strategy (lecture-aligned):** automate repetitive execution and measurement (unit/integration/perf measurement, coverage, suite exports) while restricting heavy/optional dependencies (FinBERT disabled in CI via env flag; IB not required). This maximises cost-effectiveness and visibility.

(d) **Demonstrate the CI pipeline functions as expected**

**Evidence:** green CI runs + downloadable artefacts. Local reproducibility commands are documented in `docs/how_to_run.md`.